DIMITRIS OIKONOMOU

doikono1@jh.edu | https://dimitris-oik.github.io/ | Google Scholar

RESEARCH INTERESTS

Broadly interested in large-scale optimization and machine learning. My current research focuses on adaptive step-size methods and sharpness-aware minimization, with applications to deep learning and large language model training.

EDUCATION

Johns Hopkins University — Baltimore, USA
PhD in Computer Science — *08/2023 – Present*
- Advisor: Dr. Nicolas Loizou

National Technical University of Athens — Athens, Greece
MSc in Data Science and Machine Learning — *GPA: 9.8/10*
- Thesis: Efficient Learning Algorithms of Time Varying Ranking Distributions, Advisor: Dr. Dimitris Fotakis

University of Göttingen — Göttingen, Germany
MSc in Mathematics — *GPA: 1.2 (German scale; $\approx$ 9.7/10)*
- Thesis: Convergence Results for Entropic Transfer Operators, Advisor: Dr. Bernhard Schmitzer

National and Kapodistrian University of Athens — Athens, Greece
BSc in Mathematics — *GPA: 9.6/10*

PUBLICATIONS

[6] **Dimitris Oikonomou**, Matthew Buchholz, Yuen-Man Pun, Nicolas Loizou, Robert M. Gower. *Taking the Road Less Scheduled with Adaptive Polyak Steps*. Under Submission.

[5] **Dimitris Oikonomou** and Nicolas Loizou. *Adaptive Sharpness-Aware Minimization with a Polyak-type Step Size: A Theory-Grounded Scheduler*. In **ICML 2026**

[4] **Dimitris Oikonomou** and Nicolas Loizou. *Safeguarded Stochastic Polyak Step-Sizes for Non-smooth Optimization: Robust Performance Without Small (Sub)Gradients*. In **ICML 2026**

[3] Robert Gower, Guillaume Garrigos, Nicolas Loizou, **Dimitris Oikonomou**, Konstantin Mishchenko, Fabian Schaipp. *Analysis of an Idealized Stochastic Polyak Method and its Application to Black-Box Model Distillation*. arXiv preprint arXiv:2504.01898, 2025

[2] **Dimitris Oikonomou** and Nicolas Loizou. *Sharpness-Aware Minimization: General Analysis and Improved Rates*. In **ICLR 2025**

[1] **Dimitris Oikonomou** and Nicolas Loizou. *Stochastic Polyak Step-sizes and Momentum: Convergence Guarantees and Practical Performance*. In **ICLR 2025**

TALKS & PRESENTATIONS

Invited Talk, **SIAM Conference on Optimization (OP26)**, Minisymposium	Edinburgh, 2026
Invited Talk, **ELLIIT Focus Period on Optimization for Learning**	Lund, 2026
Poster, **ELLIIT Symposium on Optimization for Learning**	Lund, 2026
Poster (x2), **International Conference on Machine Learning (ICML 2026)**	Seoul, 2026
Poster (x2), **International Conference on Learning Representations (ICLR 2025)**	Singapore, 2025

HONORS & AWARDS

International Mathematical Olympiad (**IMO**): *Bronze Medal*	Colombia, 2013
JHU MINDS Data Science Fellowship	2024, 2025
DAAD Scholarship (German Academic Exchange Service)	2019–2020
Hellenic Post Scholarship	2015–2019

TEACHING & PROFESSIONAL SERVICE

Teaching Assistant: *Computer Vision* — JHU, Fall 2024
Reviewer: ICML, NeurIPS, ICLR, AISTATS, IEEE Trans. Signal Processing, Information and Inference

TECHNICAL SKILLS

Languages & Tools: Python (PyTorch, NumPy), C/C++ (OpenMP, CUDA), R, LaTeX